LAIYU®

Laizhou city laiyu chemical co ., ltd

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Technical Data Sheet

Magnesium Sulphate Monohydrate

HS Code: 28332100

CAS #:14168-73-1

EN Number: 231-298-2

Formula: $MgSO_4 \cdot H_2O$

Molecular weight: 138.37

Shelf life: 1 year

PROPERTIES

- Appearance: powder
- Color: white
- Odor: odorless
- Specific Gravity: 0.8-0.9 g/ cm^3
- Solubility: 30g/100ml (@20°C),

SPECIFICATIONS

Item	Standard
Assay	≥ 99%
Soluble MgO	≥ 28.6%
Heavy Metals (as Pb)	≤ 0.001%
Chlorine(Cl)	≤ 0.06%
Iron(Fe)	≤ 0.002%
Arsenic (As)	≤ 0.0002%
Partical Size	Powder 40-150mesh
Sulphur (S)	≥ 22.89%
PH (5% w/v solution)	8-10

APPLICATION

- In the cosmetic industry is used in the preparation bath salts and cosmetics as it prevents ionic strength, some of the temporary skin wrinkling.
- It is also used in the treatments of aches and pains and also acts as an antiarrhythmic agent for cardiac arrest.
- In marine aquaria, it is also used to maintain the magnesium concentration.
- Magnesium sulfate heptahydrate is used to prepare copper sulfate as the electrolyte.

PACKING

- 25 Kg PP bag or 1000kg bag
- 20 MT/ 20' FCL (with pallet)
- 23 MT/ 20' FCL (without pallet).

STORAGE AND TRANSPORTATION

- Store in a cool, dry and well-ventilated place.
- Protect from extremes of temperature and humidity during storage

The above information, while not guaranteed, reflects our analysis and believed to be accurate. It is the end user's responsibility, prior to use the above mentioned product, to make his/her own determination as to the suitability of such product. D.I.S. shall not be, directly or indirectly, responsible for damages of any nature whatsoever resulting from the use or reliance upon the above mentioned information.

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name: Magnesium sulphate n-hydrate (MgSO4*nH2O)
REACH registration No.: 01-2119486789-11-0013

1.2 Relevant identified uses of the substance or mixture and uses advised against

General use: Chemical basicmaterial

Identified uses:
1. Manufacture, processing and distribution of substances and mixtures
2. Use in laboratories
3. use as an intermediate
4. Use in adhesives and sealants
5. Raw material for the fertilizer
6. Raw material for the pharmaceuticals
7. Raw material for the ink and toners
8. Raw material for the Cosmetics, personal careproducts

1.3 Details of the supplier of the safety data sheet

Company name: Laizhou City Laiyu Chemical Co.,ltd

Address.: Chenggang Road Zhuyou,Laizhou ,Shandong,China

Postal Code: 261416

Telephone: +86 535 2719337 / +86 535 2719339
Telefax: +86 535 2755678

Reach Only Representative:
Chemical Inspection and Regulation Service Limited
E-mail info@cirs-reach.com

Additional information: This safety data sheet pertains to the following products:
CAS 7487-88-9, 14168-73-1, 17830-05-6, 15320-30-6, 15553-21-6, 10034-99-8
Magnesium sulphate n-hydrate (MgSO4*nH2O)

1.4 Emergency telephone number

Chen Xue Yan, Telephone: +86 173 535 79630

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification according to EC regulation 1272/2008 (CLP)

This substance is classified as not hazardous.

2.2 Label elements

Labelling (CLP)

Hazard statements: not applicable

Precautionary statements: not applicable

2.3 Other hazards

No risks worthy of mention.

Results of PBT and vPvB assessment:

No data available

SECTION 3: Composition / information on ingredients

3.1 Substances

Chemical characterisation: Mg O4 S = MgSO4
Magnesium sulphate >= 99,0%

CAS-Number: 7487-88-9

EC-number: 231-298-2

Additional information: also as Magnesium sulphate n-hydrate (MgSO4*nH2O) >= 99,0%
CAS 14168-73-1, 17830-05-6, 15320-30-6, 15553-21-6, 10034-99-8

SECTION 4: First aid measures

4.1 Description of first aid measures

In case of inhalation: Provide fresh air.

Following skin contact: Change contaminated clothing. Remove residues with water.

After eye contact: With eyelids open, wash out eyes for several minutes under flowing water.

After swallowing: Rinse mouth and drink large quantities of water. Seek medical aid in case of troubles.

4.2 Most important symptoms and effects, both acute and delayed

Nausea, vomiting, diarrhoea.

4.3 Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media:

Product is non-combustible. Extinguishing materials should therefore be selected according to surroundings.

5.2 Special hazards arising from the substance or mixture

Fires in the immediate vicinity may cause the development of dangerous vapours.
In case of fire may be liberated: Sulphur oxides.

5.3 Advice for firefighters

Special protective equipment for firefighters:

Wear a self-contained breathing apparatus and chemical protective clothing.

Additional information: Hazchem-Code: -

Suppress gases/vapours/mists with water spray jet.
Do not allow water used to extinguish fire to enter drains, ground or waterways.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Avoid contact with the substance. Avoid generation of dust. Do not breathe dust. Provide adequate ventilation.

6.2 Environmental precautions

Do not allow to penetrate into soil, waterbodies or drains.

6.3 Methods and material for containment and cleaning up

Collect dry and place in appropriate containers for disposal. Subsequent cleaning. Avoid generation of dust.

6.4 Reference to other sections

Refer additionally to section 8 and 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advices on safe handling: Avoid contact with the substance. Avoid generation of dust. Do not breathe dust. Provide adequate ventilation, and local exhaust as needed.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storerooms and containers:

Keep container tightly closed and dry.

7.3 Specific end use(s)

No information available.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Additional information: Contains no substances with occupational exposure limit values.

DNEL/DMEL: Systemic effects:
DNEL Long-term, workers, dermal: 21.3 mg/kg bw/d
DNEL Long-term, workers, inhalative: 37.6 mg/m³
DNEL Long-term, consumers, oral: 12.8 mg/kg bw/d
DNEL Long-term, consumers, dermal: 12.8 mg/kg bw/d
DNEL Long-term, consumers, inhalative: 11.1 mg/m³

PNEC: PNEC water (freshwater): 0.68 mg/L
PNEC water (marine water): 0.068 mg/L
PNEC water (intermittent release): 6.8 mg/L
PNEC sewage treatment plant: 10 mg/L

8.2 Exposure controls

Provide adequate ventilation, and local exhaust as needed.

Personal protection equipment

Occupational exposure controls

Respiratory protection: In case of dust formation: Particulates filter P1 according to EN 143.

Hand protection: Protective gloves according to EN 374.
Glove material: Nitrile rubber-Layer thickness: 0.11 mm
Breakthrough time: > 480 min
Observe glove manufacturer's instructions concerning penetrability and breakthrough time.

Eye protection: Tightly sealed goggles according to EN 166.

Body protection: Wear suitable protective clothing.

General protection and hygiene measures:
Change contaminated clothing.
Wash hands before breaks and after work.
Work place should be equipped with a shower and an eye rinsing apparatus.
When using do not eat, drink or smoke.
Avoid contact with skin and eyes.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance:	Form: solid Colour: colourless up to white
Odour:	odourless
Odour threshold:	No data available
pH value:	approx. 600 kg/m³ at 25 °C, 50 g/L: approx. 7.9
Melting point/freezing point:	No data available
Initial boiling point and boiling range:	No data available
Flash point/flash point range:	non-flammable
Evaporation rate:	No data available
Flammability:	No data available
Explosion limits:	No data available
Vapour pressure:	low
Vapour density:	No data available
Density:	at 20 °C: 2.66 g/cm³
Water solubility:	at 0 °C: 269 g/L at 20 °C: 360 g/L at 40 °C: 456 g/L
Partition coefficient: n-octanol/water:	No data available
Auto-ignition temperature:	No data available
Decomposition temperature:	1124 °C
Viscosity, kinematic:	No data available
Explosive properties:	not explosive
Oxidizing characteristics:	not oxidising

9.2 Other information

Ignition temperature: non-flammable

Additional information: Molar mass: 120.37 g/mol

SECTION 10: Stability and reactivity

10.1 Reactivity

Refer to 10.3

10.2 Chemical stability

Product is stable under normal storage conditions.

10.3 Possibility of hazardous reactions

No hazardous reaction when handled and stored according to provisions.

10.4 Conditions to avoid

Keep away from heat.

10.5 Incompatible materials

No data available

10.6 Hazardous decomposition products

Fires in the immediate vicinity may cause the development of dangerous vapours.
In case of fire may be liberated: Sulphur oxides.

Thermal decomposition: 1124 °C

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity: LD50 Rat, oral: > 2000 mg/kg
LD50 Rat, dermal: > 2000 mg/kg

Toxicological effects:

Acute toxicity (oral): Based on available data, the classification criteria are not met.

Acute toxicity (dermal): Based on available data, the classification criteria are not met.

Acute toxicity (inhalative): Based on available data, the classification criteria are not met.

Skin corrosion/irritation: Based on available data, the classification criteria are not met.

Eye damage/irritation: Based on available data, the classification criteria are not met.

Sensitisation to the respiratory tract: Lack of data.

Skin sensitisation: Based on available data, the classification criteria are not met.

Germ cell mutagenicity/Genotoxicity: Based on available data, the classification criteria are not met.

Carcinogenicity: Based on available data, the classification criteria are not met.

Reproductive toxicity: Based on available data, the classification criteria are not met.

Effects on or via lactation: Lack of data.

Specific target organ toxicity (single exposure): Based on available data, the classification criteria are not met.

Specific target organ toxicity (repeated exposure): Based on available data, the classification criteria are not met.
NOAEL dermal: 256 mg/kg bw/d
NOAEC inhalative: 222 mg/m³
NOAEL oral: 256 mg/kg bw/d

Aspiration hazard: Lack of data.

Symptoms

In case of ingestion: After intake of large amounts: Nausea, vomiting, diarrhoea.

SECTION 12: Ecological information

12.1 Toxicity

Aquatic toxicity:

Algae toxicity:
IC50 Desmodesmus subspicatus (green algae): 2700 mg/L/72h.
Bacterial toxicity:
EC50 Photobacterium phosphoreum: 84000 mg/L/30 min.
Daphnia toxicity:
EC50 Daphnia magna (Big water flea): 1700 mg/L/24h.
Fish toxicity:
LC50 Gambusia affinis: 15500 mg/L/96h.
Source: IUCLID.

12.2. Persistence and degradability

Further details: Methods for the determination of biodegradability are not applicable to inorganic substances.

12.3 Bioaccumulative potential

Partition coefficient: n-octanol/water:
No data available

12.4 Mobility in soil

No data available

12.5 Results of PBT and vPvB assessment

No data available

12.6 Other adverse effects

General information: Do not allow to enter into ground-water, surface water or drains.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product

Waste key number: 06 03 14 = Wastes from the MFSU of salts and their solutions and metallic oxides.
Recommendation: Dispose of waste according to applicable legislation.

Contaminated packaging

Recommendation: Dispose of waste according to applicable legislation.
Non-contaminated packages may be recycled.

SECTION 14: Transport information

14.1 UN number

ADR/RID, IMDG, IATA-DGR:

not applicable

14.2 UN proper shipping name

ADR/RID, IMDG, IATA-DGR:

Not restricted

14.3 Transport hazard class(es)

ADR/RID, IMDG, IATA-DGR:

not applicable

14.4 Packing group

ADR/RID, IMDG, IATA-DGR:

not applicable

14.5 Environmental hazards

Marine pollutant: no

14.6 Special precautions for user

No dangerous good in sense of these transport regulations.

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

No data available

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

National regulations - Great Britain

Hazchem-Code: -

No data available

15.2 Chemical Safety Assessment

For this substance a chemical safety assessment has been carried out.

SECTION 16: Other information

Further information

Reason of change: General revision (Regulation (EU) No 2015/830)
Date of first version: 2/5/2011

Department issuing data sheet

Contact person: see section 1: Dept. responsible for information

For abbreviations and acronyms, see: ECHA Guidance on information requirements and chemicalsafety assessment, chapter R.20 (Table of terms and abbreviations).

The information in this data sheet has been established to our best knowledge and was up-to-date at time of revision. It does not represent a guarantee for the properties of the product described in terms of the legal warranty regulations.